BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: MRD-10TH 100523 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** MRD-10TH 100523 (A)
VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. $ax^2 + bx + c = o$ is a quadratic equation where a, b and c are real number and:

(A) $a \neq o$
(B) $c \neq o$
(C) $b \neq o$
(D) $a, b = o$

Q2. If $2^x = 1/2$ then x = ...

(A) 1
(B) o
(C) -1
(D) -2

Q3. The roots of equation $ax^2+bx+c = o$ are real and equal if b^2-4ac = ...

(A) Zero
(B) Greater than zero
(C) Less than zero
(D) Less than are equal to zero

Q4. The product of cube roots of unity is .....

(A) 0
(B) -1
(C) 1
(D) w

Q5. If 2a + 1 : 21 :: 4 : 7, then .....

(A) a = 13/2
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) a = 9/2

Q6. The third proportional of x and y is .....

(A) xy
(B) x/y
(C) y^2/x
(D) None of these

Q7. In a set of data 41, 43, 47, 51, 57, 52, 59 median is .....

(A) 51
(B) 47
(C) 57
(D) None of these

Q8. In the given data 25, 30, 31, 25, 35 the model size is .....

(A) 30
(B) 31
(C) 35
(D) 25

Q9. π/6 radians are equal to .....

(A) 60°
(B) 30°
(C) 90°
(D) 180°

Q10. A chord that contains the centre of a circle is ..... of the circle:

(A) Diameter
(B) Tangent
(C) Radius
(D) Secant

Q11. Two tangents drawn to a circle from a point outside it are of ..... in length.

(A) half
(B) equal
(C) Double
(D) triple

Q12. The measure of entire circle is .....

(A) 0°
(B) 90°
(C) 180°
(D) 360°

Q13. An angle whose vertex is on a circle and whose sides contains chords of the circle is ..... angle.

(A) inscribed
(B) escribed
(C) major
(D) acute

Q14. $1 + \tan^2\theta =$ .....

(A) $\sin^2\theta$
(B) $\cos^2\theta$
(C) $\text{cosec}^2\theta$
(D) $\sec^2\theta$

Q15. $\text{cosec}\theta$ .....

(A) $1/\sec\theta$
(B) $1/\cos\theta$
(C) $\sin\theta$
(D) $1/\sin\theta$

Section – B

Time Allowed: 2.40 Hours	**Maximum Marks:** 36

Q2. Attempt any nine (9) parts:-

(i) Solve the equation $x^2/3 - x/12 = 1/24$ using quadratic formula.

(ii) Solve $\sqrt{5x21} = x + 3$.

(iii) Evaluate $(1 + w - w^2)^7$.

(iv) If the roots of $x^2 - 7x + k = 0$ are differ by unity, then find k.

(v) Gold and silver are mixed in 7:4 if 36 grams of silver is used. How much gold is used?

(vi) Prove that $a/b = c/d = e/f = \sqrt{a^2+c^2+e^2} / b^2+d^2+f^2$

(vii) Resolve into partial fractions $(3x - 2)/(2x^2 - x)$

(viii) Prove associative property of intersection for A = {1,2,3}, B= {2,3,4} and C = {3,4,5}

(ix) If A= {1,2,3}, B= {4,5} then (i) write three binary relations from A to B.

(x) Find the standard deviation for 6,8,10,12,14

(xi) Prove that $(\cos^4 x - \sin^4 x = 1 - 2\sin^2 x$

(xii) Find the area of a sector with circular angle of 60^0 in a circular region whose radius is 5cm.

Section – C

Maximum Marks: 24

> Note: Attempt any three (3) questions:-

Q3. A light house is 150m above the sea level. Angle of depression of a boat from its top is 60^0 Find the distance between the boat and light house

Q4. Construct a triangle with sides 2cm, 2.5cm and 3cm. Also draw its circumcircle.

Q5. Prove that two tangents drawn to a circle from a point outside it, are equal in length.

Q6. The angle in a segment greater than a semicircle is less than a right angle. Prove it.